Problem 1: Stationery Shop Receipts (Units 1 & 2)

CODE

```
import java.util.Scanner;

class Product {
    private final String name;
    private final int price;

    public Product(String name, int price) {
        this.name = name;
        this.price = price;
    }

    public String getName() {
        return name;
    }

    public int getPrice() {
        return price;
    }

    public static Product findProduct(char code) {
        return switch (code) {
            case 'N' -> new Product("Notebook", 45);
            case 'P' -> new Product("Pen", 10);
            case 'M' -> new Product("Marker", 25);
            case 'F' -> new Product("File", 30);
            default -> null;
        };
    }
}

class Purchase {
    private int receiptNo;
    private Product product;
    private int quantity;

    private static int nextReceipt = 201;
    private static int totalPurchases = 0;
    private static int totalRevenue = 0;

    public Purchase(Product product, int quantity) {
        this.product = product;
        this.quantity = quantity;
        this.receiptNo = nextReceipt++;

        totalPurchases++;
```

```
        totalRevenue += getAmount();
    }

    public int getAmount() {
        return product.getPrice() * quantity;
    }

    public static int getTotalPurchases() {
        return totalPurchases;
    }

    public static int getTotalRevenue() {
        return totalRevenue;
    }

    @Override
    public String toString() {
        return "Receipt " + receiptNo + ": " + product.getName() + " x
" + quantity + " = " + getAmount();
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            char code = sc.next().charAt(0);
            int quantity = sc.nextInt();

            Product product = Product.findProduct(code);

            if (product == null || quantity <= 0) {
                System.out.println("Invalid item: " + code + " " +
quantity);
            } else {
                Purchase purchase = new Purchase(product, quantity);
                System.out.println(purchase);
            }
        }

        System.out.println("Total receipts: "
                + Purchase.getTotalPurchases()
                + ", Total revenue: "
                + Purchase.getTotalRevenue());
```

```
        sc.close();
    }
}
```

O/P

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC> javac Main.java
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC> java Main
5
N 2
Receipt 201: Notebook x 2 = 90
P 5
Receipt 202: Pen x 5 = 50
Z 1
Invalid item: Z 1
M 0
Invalid item: M 0
F 3
Receipt 203: File x 3 = 90
Total receipts: 3, Total revenue: 230
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC>
```

Problem 2: Notice Cleaner (Unit 3)

CODE

```
import java.util.Scanner;

public class String3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String sentence = sc.nextLine();
        String keyword = sc.nextLine();

        String trimmed = sentence.trim();
        String[] words = trimmed.split(" +");

        StringBuilder cleaned = new StringBuilder();

        for (int i = 0; i < words.length; i++) {
            if (i > 0) {
                cleaned.append(" ");
            }
            cleaned.append(words[i].toLowerCase());
        }

        cleaned.setCharAt(
```

```
                0,
                Character.toUpperCase(cleaned.charAt(0))
        );

        String shortestWord = words[0].toLowerCase();
        int keywordCount = 0;
        int vowelCount = 0;

        for (String word : words) {
            String lowerWord = word.toLowerCase();

            if (lowerWord.length() < shortestWord.length()) {
                shortestWord = lowerWord;
            }

            if (lowerWord.equals(keyword.toLowerCase())) {
                keywordCount++;
            }
        }

        for (int i = 0; i < sentence.length(); i++) {
            char ch = Character.toLowerCase(sentence.charAt(i));

            if (ch == 'a' || ch == 'e' || ch == 'i'|| ch == 'o' || ch
== 'u')
            {
                vowelCount++;
            }
        }

        System.out.println("Sentence: " + cleaned);
        System.out.println("Words: " + words.length);
        System.out.println("Shortest word: " + shortestWord);
        System.out.println("Keyword count: " + keywordCount);
        System.out.println("Vowels: " + vowelCount);

        sc.close();
    }
}
```

O/P

```
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC> javac String3.java
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC> java String3
 spring SEMESTER exam is   near and exam is TOUGH
EXAM
Sentence: Spring semester exam is near and exam is tough
Words: 9
Shortest word: is
Keyword count: 2
Vowels: 15
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC> java String3
The  theory is OK  the theme
the
Sentence: The theory is ok the theme
Words: 6
Shortest word: is
Keyword count: 2
Vowels: 8
PS C:\Users\Administrator\Desktop\CEUE203_OOP_25AIML072_REM_PRAC>
```